



KUWAIT UNIVERSITY
Faculty of Science
Department of Mathematics

Math 101: Calculus I

Second Exam, 18 November 2018

Duration: 90 minutes

- *In questions 1-7, show all your work.*
- *In question 8, select the correct answer (no justification is required).*
- *Calculators and communication devices are not allowed in the examination room.*

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (10 pts)	
Q2 (10 pts)	
Q3 (10 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (10 pts)	
Q7 (15 pts)	
Q8 (25 pts)	
Total	

1. [10 pts.] Evaluate the limit $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{3\theta^2}$, if it exists.

2. [10 pts.] Let $f(x) = \log_a(2x^3 - 1)$. For what value of a , if any, is $f'(1) = 3$.

3. [10 pts.] A rectangle has a fixed area of 24 cm^2 . The length is decreasing at a rate of 2 cm/min . At what rate the width is increasing when the length is 8 cm .

4. [10 pts.] Let $f(x) = 3x^2 + 4x - \ln(\cos x)$.

- (a) Find the linearization of f at $x = 0$.
- (b) Estimate $f(0.01)$.

5. [10 pts.] Find the absolute maximum and the absolute minimum values, if any, of $f(x) = \sin(2x) - 2x$ on the interval $[0, 2\pi]$.

6. [10 pts.] Find $\frac{dy}{dx}$ if $x^y = y^x$.

7. [15 pts.] Find equation of all tangent lines to the curve $5y^2 - 4xy^3 = x^2$ at $y = 1$.

8. [$5 \times 5 = 25$ pts.] In each of the following multiple choice questions, select the correct answer.

I. If $x = \ln\left(\frac{5}{2}\right)$, then $\cosh(x) =$

a) $\frac{29}{20}$.

b) $\frac{21}{20}$.

c) $\frac{20}{21}$.

d) 0.

e) None of the above.

II. The value of $\lim_{x \rightarrow 0} \frac{e^{x^2+3x} - 1}{x} =$

- a) 2.
- b) 3.
- c) 6.
- d) Does not exist.
- e) None of the above.

III. If $f(x) + x \sin(f(x)) = x^3 + \frac{\pi}{2}$, then $f'(0)$ is equal to

- a) 2.
- b) -1 .
- c) 0.
- d) 1.
- e) None of the above.

IV. The value of $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x^2 + 2x} =$

- a) 1.
- b) 0.
- c) -1 .
- d) Does not exist.
- e) None of the above.

V. Let $f(x) = \ln(x^2 + 2x + 2)$ be defined on $[-2, 0]$. One of the following statements is true

- a) 0 is the absolute maximum value of f .
- b) f has no critical numbers in $(-2, 0)$.
- c) f has an absolute minimum value at $x = -1$.
- d) $x = -\ln 2$ is a critical number of f .
- e) None of the above.